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Wide base, leaning in: how Inca dry-stone walls ride out earthquakes

The free guide to the reel on Machu Picchu's stonework: why mortar-free walls lean inward, why doors are trapezoids, why the Incas built on fault lines, and a block-stacking test you can do on a table.



A city of dry stone

Machu Picchu was built around 1450 under the Inca ruler Pachacutec. Its walls are classic Inca fine ashlar masonry: finely cut dry-stone, laid without mortar. Trapezoid shapes appear everywhere: doorways, niches, and the 'Serpent's Door' in the Temple of the Sun.

The city sits at about 2,430 m on a ridge above the Urubamba River, and it is built over earthquake faults.

Why the shape matters

A dry wall has no glue to hold it together. It is held by weight and friction between the blocks. In an earthquake, a straight tall wall has its upper blocks slide sideways until they lose contact and fall.

A wall that is wide at the base and leans inward is different: the load of the upper blocks points back into the wall, so the blocks can shift a little and settle again. Shaking-table studies of Inca dry-stone masonry treat it as a friction system: the walls resist seismic load through the friction between stones.

A 2018 study reported by Peruvian media suggests a magnitude 6.5 or greater quake hit the site around 1450, during construction, loosening blocks at the Temple of the Sun, and that the Incas kept developing trapezoid walls with large blocks at the base afterwards.

Built on a fault line, on purpose

A 2019 study suggests the builders chose fault lines because the fractured rock along them was a ready supply of building stone. The faults were the quarry.

Built examples

Temple of the Sun, Machu Picchu	A curved tower of fine ashlar with a trapezoid window and the Serpent's Door.
Lower-town doorways	Trapezoid doors, wider at the bottom than at the top.
Agricultural terraces (about 4.9 hectares)	Retaining walls and layered fill that drain the heavy rain and prevent erosion.

Try it yourself: stack blocks two ways

1. Collect 14 small flat blocks (dominoes, wooden bricks or books of similar size) and a tray.
2. Build a straight tower of 7 blocks with no glue. Shake the tray sideways, a little harder each time, and count how many shakes it survives.
3. Rebuild with the same blocks, but make each row slightly narrower than the one below (a trapezoid) and lean the sides inward.
4. Shake again. The leaning stack shifts a little and settles; the straight one slides apart.
5. Roughen the surfaces with sandpaper or paper and shake once more: more friction, more shakes survived.

The rule to remember

Don't make it rigid: give it a wide base, a lean toward its own centre, and let friction do the damping.

Default answer	Glue everything together as strong as possible.
Form-finder answer	Shape it so it can shift and settle without losing contact.
Check	Ask: if this moved a little, would it fall apart or settle back?

Go further

Machu Picchu (Wikipedia)
https://en.wikipedia.org/wiki/Machu_Picchu

Inca architecture (Wikipedia)
https://en.wikipedia.org/wiki/Inca_architecture

World History Encyclopedia: Inca ingenuity
https://etc.worldhistory.org/travel/inca-ingenuity-andes/

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Facts: Wikipedia (Machu Picchu), Peruvian Times report on the 2018 study, shaking-table research on Inca dry-stone masonry.

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